

PUEBLO, CO, USA

WMO: 724640

Lat: 38.289N

Lon: 104.506W

Elev: 4729

StdP: 12.35

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 93058

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	1.0	7.2	-7.5	4.4	17.1	-2.9	5.6	25.2	32.7	43.9	28.6	45.5	4.0	260	0.586

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	31.7	98.7	62.3	96.2	62.1	93.5	61.7	67.1	85.4	66.0	84.4	64.9	83.3	11.0	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.8	102.3	69.7	61.3	96.8	69.5	59.8	91.7	69.9	34.2	84.6	33.2	84.0	32.3	82.8	71.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.8	24.8	20.4	DB	-9.5	102.9	6.7	1.8	-14.3	104.2	-18.3	105.3	-22.0	106.3	-26.9	107.6	(4)
(5)			WB	-10.1	69.5	6.5	1.1	-14.7	70.3	-18.5	70.9	-22.2	71.5	-26.9	72.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.1	31.9	34.6	43.5	50.9	60.7	71.4	76.8	74.4	66.5	52.7	40.9	31.4		
	DBStd	18.08	10.23	10.44	9.27	8.45	7.59	6.18	4.53	4.43	7.18	8.80	9.12	10.22		
	HDD50	2283	565	435	237	90	11	0	0	0	3	72	292	578		
	HDD65	5378	1027	850	666	425	172	17	1	2	66	387	724	1041		
	CDD50	3404	2	5	36	116	343	642	832	756	498	155	17	2		
	CDD65	1025	0	0	0	1	38	209	368	293	111	5	0	0		
	CDH74	15374	1	4	57	239	1051	3292	4791	3597	1933	382	26	1		
	CDH80	7858	0	0	4	41	405	1760	2751	1896	897	103	1	0		
(14) Wind	WSAvg	7.8	6.7	7.6	8.6	9.8	8.9	8.7	8.0	7.4	7.2	7.3	6.7	6.7		
(15) Precipitation	PrecAvg	11.5	0.3	0.3	0.8	1.1	1.2	1.3	2.0	1.9	0.8	0.7	0.5	0.4		
	PrecMax	17.9	0.9	1.4	3.0	5.3	2.7	4.3	5.2	4.4	2.7	2.6	2.5	1.0		
	PrecMin	6.3	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.1	0.0	0.0	0.0		
	PrecStd	2.7	0.2	0.3	0.6	1.1	0.7	1.1	1.2	1.1	0.6	0.6	0.5	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.7	72.9	79.5	84.7	93.6	101.3	102.6	99.4	96.0	87.4	77.8	68.8		
		MCWB	45.3	46.6	49.9	52.1	58.5	61.0	63.3	63.0	61.2	55.5	49.4	45.3		
	2%	DB	63.3	66.9	74.8	80.3	88.5	97.6	99.5	96.3	92.4	82.9	72.3	62.6		
		MCWB	42.6	44.4	47.6	50.9	56.3	60.2	62.8	62.6	59.8	54.0	47.5	42.6		
	5%	DB	57.9	62.1	70.3	76.4	84.7	94.2	97.0	93.6	89.4	78.4	66.9	57.0		
		MCWB	40.2	42.2	46.1	49.7	55.6	60.1	62.9	62.7	59.0	52.2	45.5	40.1		
	10%	DB	51.8	55.5	65.1	71.9	80.3	90.6	94.2	90.7	85.8	72.9	60.8	50.9		
		MCWB	37.6	39.2	44.1	48.1	54.3	59.8	62.8	62.7	58.6	50.7	42.9	37.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.2	47.4	51.2	55.0	62.6	67.0	68.9	68.3	65.4	59.1	50.4	46.2		
		MCDB	66.9	70.2	75.7	74.8	82.1	88.0	86.2	85.3	85.3	75.8	73.9	67.0		
	2%	WB	43.2	44.9	48.9	52.8	59.9	64.7	67.5	66.9	63.3	56.4	48.2	43.1		
		MCDB	61.7	66.0	70.9	73.5	79.0	85.6	85.3	85.0	82.0	74.9	70.7	61.6		
	5%	WB	40.8	42.5	46.9	51.1	58.1	63.3	66.3	65.8	61.8	53.9	46.0	40.5		
		MCDB	56.9	60.8	68.3	71.7	76.8	83.8	84.5	83.7	81.6	73.8	65.7	56.4		
	10%	WB	38.0	39.7	44.7	49.5	56.5	62.0	65.3	64.7	60.3	51.8	43.5	37.5		
		MCDB	50.8	54.7	63.7	69.5	74.8	82.5	83.8	82.0	79.3	70.9	59.3	50.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	31.7	31.2	32.8	30.8	30.7	32.9	31.7	30.3	32.2	32.6	32.1	30.8		
		MCDBR	41.6	41.8	42.9	41.4	39.5	39.8	37.2	35.6	38.4	42.2	43.1	40.6		
	5% WB	MCWBR	23.9	22.6	20.9	17.9	14.9	13.3	10.9	10.2	13.5	18.5	22.4	23.5		
		MCWBR	39.6	40.5	40.1	36.9	33.4	32.8	29.3	29.1	31.9	36.2	40.6	38.9		
(40) Clear-Sky Solar Irradiance	taub		0.241	0.248	0.271	0.302	0.320	0.339	0.370	0.351	0.312	0.276	0.251	0.244		
		taud	2.614	2.614	2.543	2.459	2.466	2.429	2.410	2.456	2.527	2.601	2.603	2.594		
	Ebn at Noon		309	318	317	310	304	297	286	291	298	303	301	299		
		Edh at Noon	22	25	30	35	35	36	37	34	30	25	22	21		
(44) All-Sky Solar Radiation	RadAvg		884	1169	1581	1933	2149	2363	2238	2039	1756	1328	979	782		
	RadStd		53	83	112	109	152	100	91	71	90	76	59	44		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.90	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+229	+176
(47) Regional (23 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+135	+58

Nomenclature: See separate page